

Task-Relative Information Contracts: Sufficiency, Evidence, and Hierarchical Preservation in Multi-Agent Systems

Joss Armstrong*

Abstract

Multi-agent systems often know more internally than they need to expose. This paper asks what information must cross an interface if a named receiving criterion is to remain fully usable downstream. It calls such an interface a *task-relative information contract*, or *TRI contract*. For raw state X and candidate contract $T = t(X)$, the TRI deficit $D_{\text{TRI}}(T; Y) = I(Y; X \mid T)$ measures how much information about the criterion Y is lost at the interface. Zero deficit means that the full posterior $P(Y \mid X)$ is preserved. The TRI Contract Theorem identifies the smallest deterministic contract that loses none of this information. In finite discrete systems, that contract also has minimum entropy among deterministic zero-deficit contracts.

In a distributed protocol, the total TRI deficit separates exactly into information lost during reporting and information lost during aggregation. A separate finite-evidence result shows that finer representations can require more distinctions than the available observations can support, producing a resolution–evidence band under nested refinement. The same finite-evidence structure appears independently in categorical allocation–verification and reappears when a one-bit evaluation hierarchy is used to attribute learned routing preferences at multiple levels, although the two systems solve different problems. Conservative internal refinement can also add structure while preserving the same parent-facing TRI contract. The resulting framework separates what a criterion requires, where a protocol can lose it, what resolution the evidence can support, and when the same external contract survives internal composition.

Multi-agent systems are built from components that know more internally than they can, should, or need to expose externally. A model router need not reveal a model’s complete internal state to justify a route. A monitoring hierarchy need not forward every measurement to support a parent decision. The engineering question is not simply how much information a system can move, but which distinctions must cross each boundary for the receiving task to remain solvable.

Classical sufficiency asks when a representation keeps everything needed for a target. Mechanism design asks how privately informed agents can be induced to report or act strategically [1–4]. Proper scoring rules can make probabilistic reports truthful once outcomes resolve [5–7]. Multi-agent communication methods also use information bottlenecks to learn compact relay messages for downstream action [8–10].

Task-relative preservation and minimal sufficient quotients are also formulated explicitly by Bruhacs [11]. The present paper fixes a particular statistical contract: preservation of the full posterior of a named receiving criterion. It connects that contract to distributed information generation, finite-evidence support, and conservative hierarchical refinement.

Recent work asks how little information a particular loss or policy needs. Sevetlidis characterises Bayes-sufficient representations for a fixed supervised loss [12]. Sevetlidis also studies loss shift, in which changing the loss can make a previously sufficient representation inadequate even when the joint data distribution remains fixed [13]. Walsh studies policy-equivalent and

*Ericsson Ireland. ORCID: 0009-0009-3462-9679.

regret-bounded compression under a specified consequence structure, and also studies how retained support can change as recurrent policy demands change [14, 15]. The question here has a different scope. A TRI contract does not fix the later loss on Y . A zero-deficit TRI contract preserves the full posterior $P(Y | X)$, so the same interface remains usable for any later decision problem whose loss is defined on that criterion.

A companion preprint, *Task-Sufficient Contraction: Source Selection for Machine Information Interfaces*, asks a different reduction question [16]. Once a particular consumer and loss are declared, some distinctions kept by a TRI contract may no longer matter to that consumer. For the same criterion and consumer, one important nested case is

$$X \longrightarrow S_Y \longrightarrow G_C \longrightarrow B_C, \quad (1)$$

where the arrows mean deterministic coarsening, not a required processing pipeline. The present paper studies S_Y , which preserves everything X tells the receiver about Y . It asks how a distributed protocol can produce that contract, what level of detail the available evidence can support, and when the same parent-facing contract survives conservative internal refinement. Task-Sufficient Contraction asks when a declared task can remove further source distinctions before a later optimization is chosen.

These distinctions also separate two constraints that are easy to confuse. A reporting rule can make truthful reports worthwhile without creating any new observations. Incentives and evidence are therefore different resources.

The central object is a *task-relative information contract*, or *TRI contract*. Fix raw interface state X and a named receiving criterion Y . The contract states what must remain visible about X if the receiver is to retain everything X tells it about Y . The TRI Contract Theorem identifies the coarsest deterministic zero-deficit contract and, in finite discrete systems, the minimum-entropy one.

The paper then asks three further questions. First, can a distributed reporting and aggregation protocol actually produce the required contract, and where is information about Y lost when it does not? Second, can the available evidence support the level of detail demanded by the task? Third, can a component gain richer internal structure while leaving the information required by its parent unchanged?

The finite-evidence question is motivated by a recurrence across two different systems. A one-bit evaluation hierarchy can also expose the routing preferences learned at different levels. Multiplying the local routing weights along a path gives an attribution to the node at its end; finer attributions sum to their coarser counterparts [17]. This describes the router’s learned preferences, rather than a component’s intrinsic or counterfactual value. When each active attribution node requires a minimum number of observations, the same routing history imposes $K_{aqB} \leq n$. An independently developed categorical allocation–verification problem yields $Kq \leq M$. The arithmetic is elementary. What recurs is the same tension: refinement creates more distinctions that the system may need to act on, while the same finite evidence budget must be divided among them.

The paper also organises one concrete coordination architecture into four roles: Restrict, Weight, Commit and Aggregate. These roles are not the definition of a TRI contract. Restrict is the information-contract layer. The other three describe one way to evaluate reports, collect information about the criterion, and combine the resulting consequences after the interface question has been settled. The worked architecture draws on a one-bit hierarchical evaluation interface [18, 19], a finite service-orchestration problem [20], and the multi-resolution attribution construction [17].

The scope is deliberately static. The theorems below preserve a TRI contract for a fixed receiving criterion and dependence structure. If action changes the learner, state distribution, criterion, action menu, source distribution, or dependence structure, a contract that is valid now may not be valid for the next decision. This paper does not compare TRI contracts across such

changing problems, determine whether the required interface should become finer or coarser, or define a dynamic update rule. Those questions are left outside the present static theory.

Results

The minimum task-relative information contract

Let X denote the information available at an interface and let Y be the named criterion on which the receiving component acts. Assume regular conditional distributions exist and the information quantities used below are well defined.

Definition 1 (Task-Relative Information Contract). A *task-relative information contract* (TRI contract) for criterion Y is a deterministic interface $T = t(X)$ declared for that criterion. Its TRI deficit is

$$D_{\text{TRI}}(T; Y) := I(Y; X | T). \quad (2)$$

A TRI contract is *zero-deficit* when $D_{\text{TRI}}(T; Y) = 0$, equivalently when $Y \perp X | T$. Here “contract” means an information-interface specification, not a legal or payment contract.

A canonical zero-deficit representative is the posterior map

$$S_Y(X) := P(Y \in \cdot | X), \quad (3)$$

understood up to almost-sure equivalence or relabelling. Thus $P(Y | X) = P(Y | S_Y)$ almost surely.

Theorem 1 (TRI Contract Theorem: minimal zero-deficit interface). *Let $S_Y = S_Y(X)$ be the posterior contract defined above. Then:*

1. $P(Y | X) = P(Y | S_Y)$ and $I(X; Y) = I(S_Y; Y)$.
2. For any action space and decision loss for which the Bayes risks are well defined, the infimal Bayes risk attainable from X equals the infimal Bayes risk attainable from S_Y :

$$\mathcal{R}_X^* = \mathcal{R}_{S_Y}^*. \quad (4)$$

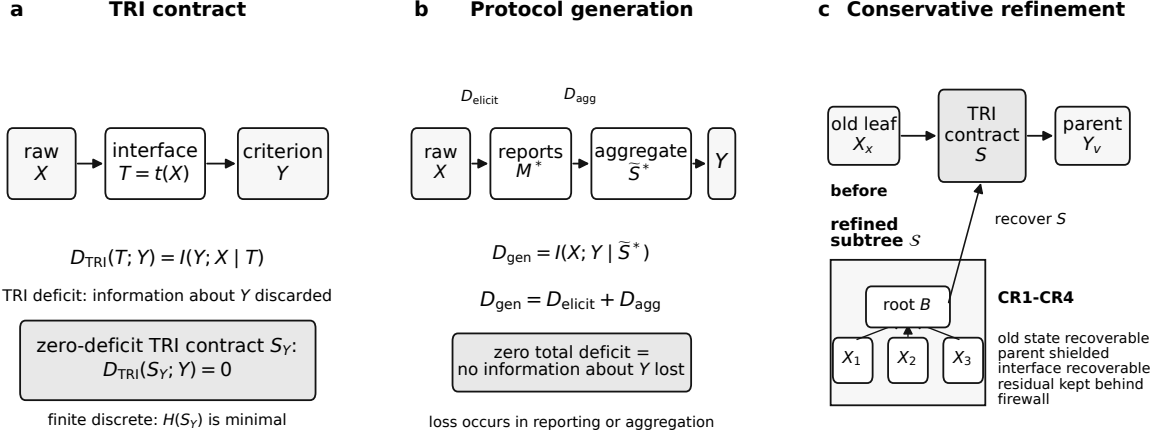
3. If $T = t(X)$ is any deterministic zero-deficit TRI contract, then $S_Y = h(T)$ almost surely for some measurable h . In a finite discrete model, $H(S_Y) \leq H(T)$.

The statistical sufficiency and decision-equivalence ingredients of Theorem 1 are classical. The theorem name refers to their role at an information interface, not to the discovery of sufficiency itself [21, 22]. A fixed-loss Bayes quotient may be smaller because it needs only enough information to choose the Bayes-optimal action [12]. The TRI contract S_Y keeps the full posterior of Y , so it remains sufficient when the later loss on Y changes under the same joint law, in contrast to the loss-specific failures examined in Sevetlidis [13]. The new questions here concern how such a contract is produced, what resolution finite evidence can support, and when the same parent-facing contract survives hierarchical refinement.

In a finite discrete model, $H(S_Y)$ is the minimum entropy among deterministic zero-deficit TRI contracts. The residual source entropy $H(X | S_Y)$ may be positive even when the contract has zero deficit. It measures information left outside the minimum contract, not a failure of sufficiency.

Because T is a function of X , its TRI deficit satisfies

$$D_{\text{TRI}}(T; Y) = I(Y; X) - I(Y; T) \geq 0, \quad (5)$$



Internal complexity may grow while the same parent-facing TRI contract is preserved.

Figure 1: **Task-relative information contracts.** **a**, A candidate TRI contract T loses information about Y by $D_{\text{TRI}}(T; Y)$. The canonical contract S_Y has zero deficit and, in finite discrete systems, minimum entropy among deterministic zero-deficit contracts. **b**, In a distributed protocol the total TRI deficit separates into loss during reporting and loss during aggregation. **c**, Under conservative refinement conditions CR1–CR4, richer internal structure can preserve the same parent-facing sufficient interface.

and $D_{\text{TRI}}(T; Y) = 0$ exactly when T is sufficient. When Y is finite, under logarithmic loss the TRI deficit equals the increase in optimal predictive risk:

$$\mathcal{R}_{\log}^*(T) - \mathcal{R}_{\log}^*(X) = H(Y | T) - H(Y | X) = D_{\text{TRI}}(T; Y). \quad (6)$$

Thus what matters is the information about the named criterion that the interface discards, not how much unrelated source entropy remains outside it.

The same distinction can matter strategically. If $X = (S, R)$ and S is sufficient, then R adds no information about Y once S is known. Exposing R may still give a strategic intermediary extra degrees of freedom that can affect behaviour. Any resulting cost depends on the mechanism. The theorem does not identify $H(R | S)$ with a universal measure of strategic vulnerability.

How a distributed protocol produces the contract

The TRI Contract Theorem identifies what the interface must preserve. It does not guarantee that distributed agents will actually produce that information. Let X now denote the complete distributed information, let $M^* = \{m_i^*\}_i$ denote the joint equilibrium report tuple, and let

$$\tilde{S}^* = \mathcal{P}(M^*) \quad (7)$$

be the protocol output. Assume the mutual informations below are finite, any protocol randomisation is independent of Y conditional on X , and $\tilde{S}^*$ is generated from M^* alone. The protocol then induces the Markov chain

$$Y - X - M^* - \tilde{S}^*. \quad (8)$$

Two things can go wrong before the receiver sees the final interface. First, the reports themselves may omit information about Y . To rule this out, the reports must preserve all information about Y present in the distributed state,

$$I(M^*; Y) = I(X; Y), \quad (9)$$

or equivalently $Y \perp X \mid M^*$. Second, the reports may contain the required information but the aggregator may discard some of it. To rule this out, aggregation must preserve the posterior,

$$I(\tilde{S}^*; Y) = I(M^*; Y), \quad (10)$$

or equivalently $Y \perp M^* \mid \tilde{S}^*$.

Theorem 2 (Contract Generation Decomposition: reporting loss and aggregation loss). *Define*

$$D_{\text{elicit}} := I(X; Y \mid M^*), \quad (11)$$

$$D_{\text{agg}} := I(M^*; Y \mid \tilde{S}^*), \quad (12)$$

$$D_{\text{gen}} := I(X; Y \mid \tilde{S}^*). \quad (13)$$

Then

$$\boxed{D_{\text{gen}} = D_{\text{elicit}} + D_{\text{agg}} \geq 0.} \quad (14)$$

Moreover, $D_{\text{elicit}} = 0$ exactly when (9) holds, $D_{\text{agg}} = 0$ exactly when (10) holds, and $D_{\text{gen}} = 0$ exactly when the protocol output is sufficient for Y relative to X .

Once the reporting Markov structure is fixed, the identity in Theorem 2 follows from the chain rule. It shows where the loss occurs. If $D_{\text{elicit}} > 0$, information is lost before or during reporting. If $D_{\text{agg}} > 0$, the reports contain information about Y that the aggregator throws away. When Y is finite, under logarithmic loss the same terms are exact increments in optimal predictive risk:

$$D_{\text{elicit}} = H(Y \mid M^*) - H(Y \mid X), \quad (15)$$

$$D_{\text{agg}} = H(Y \mid \tilde{S}^*) - H(Y \mid M^*), \quad (16)$$

$$D_{\text{gen}} = H(Y \mid \tilde{S}^*) - H(Y \mid X). \quad (17)$$

The remaining source entropy $H(X \mid \tilde{S}^*)$ need not vanish.

The decomposition does not require one particular incentive mechanism. One exact construction is available when all reporters predict the same finite criterion Y , their evidence blocks are conditionally independent given Y , reports are committed before Y is observed, and the outcome is later available for scoring. In the reporting subgame used here, a reporter can change only its report and its per-round payoff is a strictly increasing affine function of a strictly proper score. Under those assumptions, truthful posterior reporting is the unique per-round best response. If reporter i submits $q_i(y) = P(Y = y \mid X_i)$, then the prior-corrected pool

$$\bigoplus_P(q_1, \dots, q_k)(y) = \frac{P(y)^{1-k} \prod_j q_j(y)}{\sum_{y'} P(y')^{1-k} \prod_j q_j(y')} \quad (18)$$

recovers the joint posterior. Pooling within subtrees and then pooling the subtree posteriors returns the same $P(Y \mid X_1, \dots, X_n)$ as pooling all reports at once. This construction works under the stated assumptions. It is not claimed to be the only mechanism that can satisfy Theorem 2.

Finite evidence limits supported resolution

Sufficiency identifies which distinctions matter for a fixed task. It does not make evidence unlimited. Consider a finite evidence set $\Omega = \{\omega_1, \dots, \omega_M\}$ routed through a hierarchy. Each primitive observation follows exactly one root-to-leaf evidential path. For a node v , let N_v be the number of primitive observations whose path contains v . At each internal node, child evidence sets form a disjoint partition of the parent evidence set, so

$$N_v = \sum_{u \in C(v)} N_u. \quad (19)$$

For every cut C that intersects each root-to-leaf path exactly once,

$$\boxed{\sum_{v \in C} N_v = M.} \quad (20)$$

Now let $C \prec C'$ mean that C' is a finer representation than C . Assume the finer action class can reproduce every coarser action, $\mathcal{A}_C \subseteq \mathcal{A}_{C'}$, and define the best task loss

$$L(C) = \inf_{a \in \mathcal{A}_C} \ell(a). \quad (21)$$

Then the finer representation cannot worsen the best attainable task loss:

$$L(C') \leq L(C). \quad (22)$$

This is a possibility result. It does not say that finite-sample estimation improves with depth.

Suppose each active distinction requires at least $q > 0$ supporting primitive observations. A cut is evidence-feasible when $N_v \geq q$ for every $v \in C$. By (20), any such cut satisfies $|C|q \leq M$. Let J_C denote the cut-node label reached by a uniformly sampled primitive observation, so $\mathbb{P}(J_C = v) = N_v/M$. Every evidence-feasible cut therefore satisfies

$$\boxed{H(J_C) \leq \log_2 |C| \leq \log_2 \left\lceil \frac{M}{q} \right\rceil.} \quad (23)$$

This bounds the entropy of the cut label that the available evidence can support. It is necessary, not sufficient: staying below the bound does not by itself guarantee that every active node has q observations or that the representation meets the task requirement.

For a nested chain $C_0 \prec \dots \prec C_D$, assume the defining sets below are non-empty and define

$$d_R := \min\{d : L(C_d) \leq \epsilon\}, \quad (24)$$

$$d_E := \max\{d : \min_{v \in C_d} N_v \geq q\}. \quad (25)$$

If either defining set is empty, no depth satisfies the corresponding requirement. If $d_R > d_E$, the jointly feasible set is empty.

Theorem 3 (Finite-Evidence Resolution Theorem). *Under the assumptions above, the resolutions that simultaneously meet the representation tolerance ϵ and the common evidential support requirement q are exactly*

$$\boxed{d_R \leq d \leq d_E.} \quad (26)$$

The proof is elementary: monotonicity comes from action-set inclusion and the upper bound comes from conservation of a finite support budget. The theorem does not derive the support threshold q , claim that every hierarchy admits a single scalar depth ordering, or assert improved finite-sample estimation under refinement. It states exactly which depths can meet both requirements under the stated nested-refinement and evidence-accounting assumptions.

The two sides of the band come from different resources. The task sets the minimum required resolution d_R . Finite evidence sets the maximum supported resolution d_E and, through (23), an upper bound on the entropy of the distinction label. The finite-evidence theorem alone does not give a general lower entropy bound. Such a bound needs additional source and loss structure. The categorical allocation–verification example below supplies one such lower bound. Refinement creates distinctions and weakly enlarges the available action class. It does not create evidence. Evidence already attached to a coarser component is divided among the finer components that replace it. The result can therefore be summarised as: *resolution creates distinctions. It does not create evidence.*

A finite-evidence tree is closed under conservative refinement as well, but path continuation and boundary emulation are not sufficient by themselves. Suppose both the original tree and the inserted finite subtree satisfy the one-path evidence rule, local evidence conservation, nested representational refinement, action-set inclusion, and the same finite local support requirement q . If every observation formerly ending at the replaced leaf continues along exactly one new evidential path and every parent-visible action formerly available at that leaf can still be emulated at the substitution boundary, then the composite satisfies the same structural properties and therefore admits another band of the form (26).

Corollary 1 (Finite-Evidence Conservative Tree Closure). *If the original and inserted trees both satisfy the finite-evidence class assumptions above, every former observation continues through exactly one path in the inserted subtree, and the substitution boundary can emulate every formerly available parent-visible action, then the composite tree satisfies the same class assumptions and admits a resolution-evidence band.*

Conservative refinement can preserve a TRI contract

The preceding finite-evidence closure concerns counts and action classes. A stronger question is whether a sufficient parent-facing interface itself survives when a component is opened into a richer internal hierarchy. The answer is not automatic. Arbitrary replacement can alter dependence among siblings and invalidate interfaces that were sufficient only under the old dependence structure. A minimal counterexample is given in Methods. Closure therefore requires a conservative refinement operation rather than unrestricted substitution.

Let leaf x of parent v expose external state X_x through a parent-facing interface $S_{x \rightarrow v} = s_x(X_x)$. Replace the internal implementation by a subtree $\mathcal{S}$ with richer raw state $X_{\mathcal{S}}$. Let X_{-x} denote the unchanged sibling raw states and Z_{-x} their parent-facing interfaces. Four conditions are required:

1. **External-state recoverability (CR1)**. There is a deterministic map h with $X_x = h(X_{\mathcal{S}})$.
2. **Parent shielding (CR2)**. New internal detail adds no parent-relevant information once the old external state, siblings, and parent context are known:

$$Y_v \perp X_{\mathcal{S}} \mid (X_x, X_{-x}, C_v). \quad (27)$$

3. **Interface recoverability (CR3)**. The subtree root export $B_{\mathcal{S} \rightarrow v}$ permits reconstruction of the old interface: $S_{x \rightarrow v} = g(B_{\mathcal{S} \rightarrow v}, C_v)$.
4. **Interface firewall (CR4)**. The parent continues to coordinate on $S_{x \rightarrow v}$, or an equivalent relabelling, rather than silently using richer residual detail exposed by the new subtree.

Theorem 4 (Conservative TRI Contract Refinement Theorem). *Suppose the old parent interface tuple is sufficient,*

$$Y_v \perp (X_x, X_{-x}) \mid (S_{x \rightarrow v}, Z_{-x}, C_v). \quad (28)$$

If leaf x is replaced by a subtree satisfying CR1–CR4, then

$$\boxed{Y_v \perp (X_{\mathcal{S}}, X_{-x}) \mid (S_{x \rightarrow v}, Z_{-x}, C_v).} \quad (29)$$

Thus the old parent sufficiency relation survives the internal refinement.

The theorem separates internal complexity from what the parent must receive. A subsystem may gain new nodes, richer raw state, local criteria, and additional internal coordination without increasing the information its parent needs. What must stay unchanged is the receiving parent's TRI contract.

This statement is deliberately narrower than arbitrary recursive closure. If the parent changes what it values, an interface sufficient under the old criterion need not remain sufficient. For example, an interface sufficient for throughput alone may legitimately discard energy consequences. If a new intent makes energy relevant, the old interface must be revalidated or replaced by one that also preserves the newly relevant energy consequences. This is not a failure of Theorem 4. The parent-facing TRI contract has changed.

Preserving the full Restrict–Weight–Commit–Aggregate architecture requires more than the information theorem. The inserted subtree must also remain valid internally, and the parent-facing reporting, Weight rule, and decision rule must stay within the scope of the original architecture. Those conditions are stated after the four layers are defined below.

One four-layer coordination architecture

The following four roles organise one worked architecture. They are not the definition of a TRI contract and do not add another universal theorem.

Restrict. What information may cross the boundary without losing attainable decision value? The TRI Contract Theorem answers this through sufficiency and the TRI deficit D_{TRI} . The answer depends on the task. Information beyond a sufficient interface adds no Bayes decision value for the named criterion, but keeping it is not itself a loss.

Weight. How should repeated performance affect standing? In the bounded asymmetric EWMA rule used in the existing architecture, let $0 < \alpha_+, \alpha_- \leq 1$ be the update rates after correct and incorrect outcomes. Assume the conditional probability of correctness is a state-independent $p \in (0, 1)$. If trusting an incorrect report costs L_+ and rejecting a correct report costs L_- , the unique ratio that makes the *stationary mean* reputation threshold at $1/2$ coincide with the Bayes decision boundary for every such p is

$$\boxed{\rho^* = \frac{\alpha_-}{\alpha_+} = \frac{L_+}{L_-}}. \quad (30)$$

This is a local evaluator-loss calibration result. It is not the source of truthfulness in the reporting mechanism.

Commit. How does information about the criterion enter the protocol? Where outcomes later resolve, strictly proper scoring can make truthful probabilistic reporting the unique per-round best response when deviations are confined to the report and the reporter’s per-round payoff is a strictly increasing affine function of the score. The Contract Generation Decomposition then asks whether the resulting reports preserve everything they should about Y and whether the aggregator keeps that information. The canonical prior-corrected pool in (18) is one exact construction under conditional independence. Within this reporting subgame, truthful reporting needs no additional payment or verification layer beyond the scoring rule. That removes one incentive problem, but it does not create evidence or relax the finite-evidence ceiling.

Aggregate. How are parent-facing consequences converted into a collective decision? The worked architecture uses the reputation-weighted Nash social-welfare rule implemented in TACIT [23], building on the Nash social-welfare objective [24, 25]. This is one concrete aggregation rule, not a requirement of a TRI contract. At node v , with positive contextual utilities $u_{i \rightarrow v}(p; C_v)$ and fixed non-negative reputations during candidate comparison,

$$W_v(p; C_v) = \prod_i u_{i \rightarrow v}(p; C_v)^{R_{i \rightarrow v}}. \quad (31)$$

This decision aggregation is distinct from belief pooling in Commit. Positive rescaling of a child utility by a context-dependent factor constant across candidate actions leaves the ranking unchanged, so levels need not share physical KPI units or internal welfare functions.

When systems are composed, a parent need not inherit a child's internal choices. It needs only an interface sufficient for its own criterion, after which it can apply its own evaluation and decision rule.

For the theorem below, call a node locally architecture-admissible when its receiving interface is sufficient for its local criterion, its Weight rule lies inside the one-axis calibration class, its reports and aggregator preserve the required information, and its Aggregate rule uses positive contextual utilities with fixed reputation weights.

Theorem 5 (Conservative Architecture Refinement Theorem). *Let a finite tree be locally architecture-admissible at every node, and replace a leaf by an internally architecture-admissible subtree. Suppose CR1–CR4 hold and, at the receiving parent, (i) the former commitment/report protocol is reapplied to the preserved interface with the refined boundary in the same strategic scope and the parent report aggregator is unchanged, (ii) Weight error semantics remain functions of the preserved interface, realised parent criterion, and local context, and (iii) the child's Layer-4 utility contribution remains a function of that preserved interface and parent context while the parent Aggregate operator is unchanged. Then the composite tree is again architecture-admissible. By induction, the class is closed under repeated conservative interface-preserving hierarchical refinement.*

The theorem does not require child and parent levels to share a criterion, context labels, numerical loss ratio, or internal welfare function. It does require the receiving parent to see the same kind of valid boundary contract. If the meaning of the boundary changes, the receiving node must revalidate the contract rather than assume that the refinement result still applies.

Worked interfaces and finite-evidence recurrence

The first worked example is a hierarchical component-selection mechanism in which only the root observes outcome feedback [18]. Each selector maintains weights over its children. On the active path, the sign of a child's own weight change is readable locally as a one-bit evaluation signal. For the active-node event $A_{v,t}$, the earlier result gives $s_v(t) = o_t$ and hence

$$I(o_t; s_v(t) \mid A_{v,t}) = H(o_t \mid A_{v,t}). \quad (32)$$

Conditional on the sign, the pre-round state, and $A_{v,t}$, the update magnitude carries no additional information about o_t . Thus the one-bit sufficiency claim applies only to the active-path case covered by the component proof, not to every node and round. The component experiments report how accurately this signal is preserved through the tested hierarchy depths and branching factors.

The service allocation–verification study asks how one categorical interface can support both allocation and later verification [20]. Finer categories can reduce allocation error because they express more heterogeneous requirements. The same refinement divides a finite contributor population across more category-specific monitoring pools. The resulting lower and upper information bounds are

$$R_P(\epsilon) \leq H(C) \leq \log_2 \left\lceil \frac{M}{q} \right\rceil, \quad (33)$$

where $R_P(\epsilon)$ is the minimum declaration entropy needed to meet the allocation target for the stated source, and q is the minimum number of contributors per category required by the monitoring target. For the explicit finite quadratic population the stronger cardinality result is exact: a design satisfies both requirements if and only if $K_W(\epsilon) \leq K \leq \lfloor M/q \rfloor$. The designs

that attain this band have zero declaration gain, and imposing that condition does not reduce the optimal welfare in the aligned fixed-menu class. Because arbitrary reassignment can change category occupancies, the component proof constructs a design that satisfies self-selection and monitoring at the same time. This upper bound therefore comes from the available evidence, not from incentives: exact self-selection does not create more contributors. This is a concrete instance of Theorem 3. Where an aggregate category record is sufficient for the named detection criterion, the TRI Contract Theorem additionally says that the full individual record has no greater attainable Bayes decision value for that criterion.

The finite-evidence constraint reappears after composition

The two worked interfaces are not merely parallel examples. Reading the hierarchical one-bit evaluator’s routing weights as node attributions gives an attribution hierarchy that can be read at several resolutions. Each routing outcome follows one root-to-leaf path, and a complete cut partitions the same outcome history among the attribution nodes at that depth [17, 18]. Let C_d be such a complete cut, $K_d = |C_d|$, and let $N_v(n)$ count the first n primitive routing outcomes whose path crosses node v . Then

$$\sum_{v \in C_d} N_v(n) = n. \quad (34)$$

If reliable local attribution at every active node requires at least q_B observations, any supported depth must therefore satisfy

$$\boxed{K_d q_B \leq n.} \quad (35)$$

This is not an empirical curve fit. Each routing outcome crosses exactly one node in a complete cut, so the node counts must sum to n . A model of the local routing-state estimates is needed to derive the numerical value of q_B for a chosen fidelity target, but not the bound once that local requirement is fixed.

The categorical allocation–verification construction reaches the same finite-support form from a different problem. The arithmetic inequality itself is elementary counting. The point of the comparison is that hierarchical composition regenerates the same resolution-versus-evidence structure in a different coordination problem. There, K service categories share a population of M contributors, and a monitoring target requiring at least q contributors per occupied category gives

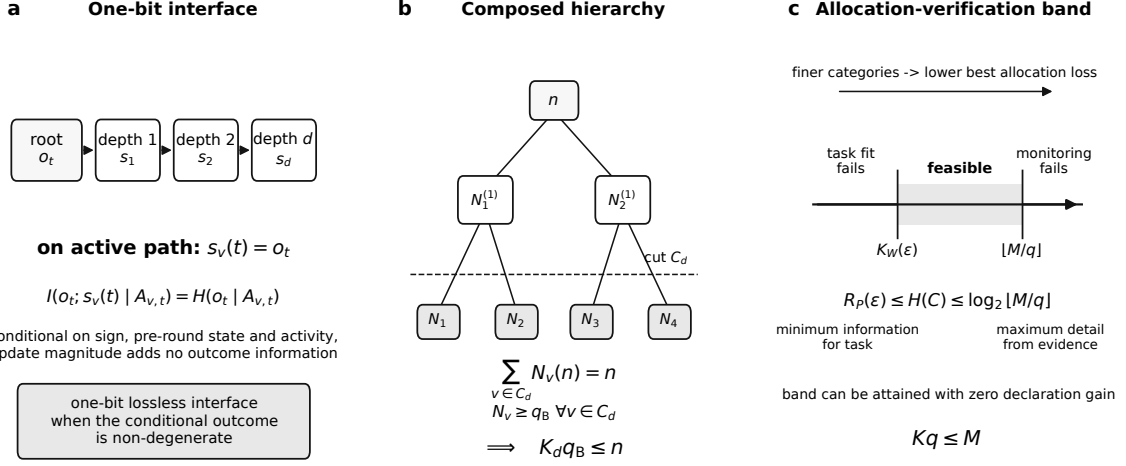
$$\boxed{Kq \leq M.} \quad (36)$$

The correspondence

$$(K_d, n, q_B) \longleftrightarrow (K, M, q) \quad (37)$$

concerns the form of the constraint, not the meaning of the variables. The attribution construction concerns resolution in an adaptive hierarchy, while the service construction concerns allocation and verification through a categorical interface. In both, refinement creates more distinctions that may matter to the task without creating more primitive evidence. When the nested cuts of the attribution hierarchy also satisfy the representation/action monotonicity assumed in Theorem 3, the same lower-resolution/upper-evidence logic gives a resolution–evidence band. The service construction goes further in its own finite quadratic setting by proving the exact existence band $K_W(\epsilon) \leq K \leq \lfloor M/q \rfloor$.

This recurrence is what motivated the general TRI-contract framework. A one-bit evaluation hierarchy can itself become a higher-level attribution problem, at which point the same resolution-versus-evidence structure appears again independently of the allocation–verification setting. The two mechanisms are not being identified with one another. They have different meanings, losses, and local estimators. The finite-evidence theorem states only the structure they share. The conservative-refinement results then state the additional conditions needed for that structure to survive further hierarchical refinement.



Same finite-evidence structure after composition: $(K_d, n, q_B) \leftrightarrow (K, M, q)$. The two systems solve different problems.

Figure 2: **Finite-evidence recurrence under composition.** **a**, In the hierarchical one-bit evaluation construction, the sign of a selected child’s weight change equals the root outcome on the active path, giving a minimal binary lossless signal when the conditional outcome is non-degenerate. **b**, Reading the same routing weights as node attributions exposes routing preferences at several resolutions. At a depth- d cut C_d , the n routing outcomes are partitioned across the cut, so $\sum_{v \in C_d} N_v(n) = n$. If every active node requires at least q_B observations, then $K_d q_B \leq n$. **c**, Independently, the finite categorical allocation–verification construction has the exact band $K_W(\epsilon) \leq K \leq \lfloor M/q \rfloor$ and the information bounds $R_P(\epsilon) \leq H(C) \leq \log_2 \lfloor M/q \rfloor$. Designs attaining the band can have zero declaration gain. Panels b and c therefore show the same finite-evidence structure in two different mechanisms. The component results motivate the general contract and finite-evidence results. They do not by themselves prove the general refinement theorems.

Discussion

The framework separates four statements that are easy to confuse. First, a receiving task can have a minimum zero-deficit contract. In a finite discrete system, the canonical contract S_Y has no more entropy than any other deterministic zero-deficit contract. Raw distinctions outside it add no Bayes decision value for that named criterion. Second, a distributed system must actually produce that contract. The generation decomposition shows whether information is lost in reporting, aggregation, or both. Third, finite evidence limits how much detail can be supported. If every active distinction requires q observations from M total, the entropy of the distinction label cannot exceed $\log_2 \lfloor M/q \rfloor$. Fourth, making a component more complex internally need not increase the information its parent must receive. Conservative refinement can preserve the former parent-facing contract exactly. The worked attribution hierarchy connects the third and fourth statements. Once the one-bit hierarchy is read as an attribution hierarchy, the same n routing outcomes must be divided among the nodes in a complete cut. This gives $K_d q_B \leq n$, matching the form $Kq \leq M$ obtained independently in categorical allocation–verification. The recurrence is not a claim that the two applications are identical. It shows that the same finite-evidence constraint can reappear when a coordination mechanism is itself embedded in a higher-level hierarchy.

The information requirement, the amount of evidence, and the incentive problem are separate. Under the scoped proper-scoring construction above, truthful reporting can be a best response without an additional payment or verification layer. Separately, the categorical allocation–

verification construction can attain its feasible band with zero declaration gain and no loss of optimal welfare in the aligned fixed-menu class. Neither result creates more observations. The evidence limit remains.

The conclusions are therefore narrower than several tempting slogans. More information is not universally harmful. A contract can be sufficient even when source information remains outside it. Incentive compatibility does not create evidence. Proper scoring does not guarantee that arbitrary reports collectively preserve everything needed about Y . A subsystem that is valid on its own can still break the parent contract when substituted into a larger system.

The practical rule is simple. Define the receiving criterion. Identify the minimum interface that must preserve it. Check that the reporting and aggregation path actually carries that information. Then ask whether the available evidence can support the required level of detail. If the task requires more detail than the evidence can support, no interface in the stated class satisfies both demands. If a component is refined internally, keep the parent-facing contract unchanged unless the parent explicitly revalidates a richer one.

The final condition marks the boundary of the present paper. The results are stated for one fixed receiving task. If action changes the learner, state population, future criterion, action menu, source distribution, or dependence structure, a TRI contract valid before the action may no longer be valid for the next decision. The present results do not compare the required contracts across those states or say whether resolution should increase or decrease. They establish the static contract and the conditions for preserving it under conservative internal refinement, not a rule for updating it after the problem itself changes.

Methods

Proof of the TRI Contract Theorem

Part (i) is immediate from sufficiency. Since S_Y is a function of X , $P(Y | X, S_Y) = P(Y | X)$. Conditional independence gives $P(Y | X, S_Y) = P(Y | S_Y)$, hence $P(Y | X) = P(Y | S_Y)$ and the mutual informations agree.

For part (ii), after observing x the full conditional risk function over actions depends on x only through $P(Y | X = x)$. By part (i) this posterior is determined by S_Y . Any Bayes-optimal action selectable from X can therefore be selected as a function of S_Y alone. Since every S_Y -measurable rule is also X -measurable, the two infima coincide.

For part (iii), if $T = t(X)$ is a zero-deficit TRI contract then $P(Y | X) = P(Y | T)$. The posterior-equivalence class represented by S_Y is therefore determined by T , so $S_Y = h(T)$ almost surely. In finite discrete systems entropy cannot increase under a deterministic map, hence $H(S_Y) \leq H(T)$.

TRI deficit and log-loss identity

For deterministic $T = t(X)$, the Markov chain $Y - X - T$ and the chain rule give

$$I(Y; X) = I(Y; T) + I(Y; X | T). \quad (38)$$

This proves non-negativity and the equivalence between $D_{\text{TRI}}(T; Y) = 0$ and sufficiency. When Y is finite, under logarithmic loss the Bayes-optimal predictive risk from information U is $H(Y | U)$, so

$$H(Y | T) - H(Y | X) = I(Y; X | T) = D_{\text{TRI}}(T; Y). \quad (39)$$

Proof of the contract-generation decomposition

From $Y - X - M^*$,

$$D_{\text{elicit}} = I(X; Y) - I(M^*; Y). \quad (40)$$

From $Y - M^* - \tilde{S}^*$,

$$D_{\text{agg}} = I(M^*; Y) - I(\tilde{S}^*; Y). \quad (41)$$

Adding the identities gives

$$D_{\text{elicit}} + D_{\text{agg}} = I(X; Y) - I(\tilde{S}^*; Y) = I(X; Y \mid \tilde{S}^*) = D_{\text{gen}}. \quad (42)$$

Each term is a conditional mutual information and is non-negative. Vanishing of the corresponding term is equivalent to the corresponding conditional independence.

Exact recursive pooling in the canonical construction

Assume positive prior support and conditional independence of disjoint primitive evidence blocks given the common criterion Y . For each reporter, Bayes' rule gives $q_i(y) \propto P(y)P(X_i \mid Y = y)$. For a non-empty group G ,

$$P(y)^{1-|G|} \prod_{i \in G} q_i(y) \quad (43)$$

is proportional to $P(y) \prod_{i \in G} P(X_i \mid Y = y) = P(y)P(X_G \mid Y = y)$, which is proportional to $P(Y = y \mid X_G)$. Thus the prior-corrected pool returns the exact group posterior. Disjoint subtrees remain conditionally independent given Y , so induction over the tree proves equality between hierarchical and flat pooling.

Proof of the finite-evidence band

Every primitive observation intersects a cut exactly once, so the node evidence sets on a cut form a disjoint partition of Ω and sum to M . If each node requires at least q observations, evidence feasibility implies $|C|q \leq M$. Along a nested refinement chain, action-set inclusion makes $L(C_d)$ non-increasing, so every depth $d \geq d_R$ meets the representation tolerance. Evidence conservation makes support feasibility downward-closed, so every $d \leq d_E$ meets the support requirement. Their intersection is exactly $d_R \leq d \leq d_E$ when $d_R \leq d_E$, and is empty otherwise.

For the finite-evidence tree result, assume both the old and inserted trees already satisfy the one-path, local-conservation, nested-refinement, action-inclusion, and common- q properties. Former observations ending at the replaced leaf are continued along exactly one new evidential path, preserving the accounting at the boundary. Boundary action emulation preserves action-set inclusion across the substitution. The composite therefore satisfies the same finite-evidence class assumptions and admits another resolution-evidence band.

Why arbitrary substitution is not closed

Let two old children have raw states A and B with $A = B$ almost surely, and let the parent criterion be $Y = A$. Suppose child A exports the constant interface $S_A = 0$ while child B exports $S_B = B$. The tuple $(0, B)$ is sufficient for Y only because of the old dependence $A = B$. Now replace child B by an otherwise internally valid subsystem with external state B' that is independent of A , exporting $S_{B'} = B'$. The new tuple $(0, B')$ contains no information about $Y = A$. Validating the inserted subsystem in isolation is therefore insufficient: substitution can invalidate an unchanged sibling interface by changing the dependence structure. CR1–CR4 rule out precisely this failure mode for conservative refinement.

Proof of conservative hierarchical sufficiency closure

By CR2,

$$P(Y_v \mid X_S, X_{-x}, C_v) = P(Y_v \mid X_x, X_{-x}, C_v). \quad (44)$$

Old parent sufficiency gives

$$P(Y_v \mid X_x, X_{-x}, C_v) = P(Y_v \mid S_{x \rightarrow v}, Z_{-x}, C_v). \quad (45)$$

CR3 makes the old interface recoverable from the subtree root and CR4 keeps the parent decision channel on that interface. Combining the equalities proves (29).

One-axis evaluator-loss calibration

Let $p \in (0, 1)$ and assume the conditional probability of a correct outcome is p independently of the current reputation state. Let $0 < \alpha_+, \alpha_- \leq 1$. Under asymmetric EWMA reputation,

$$R_{t+1} = \begin{cases} (1 - \alpha_+)R_t + \alpha_+, & \text{correct,} \\ (1 - \alpha_-)R_t, & \text{incorrect,} \end{cases} \quad (46)$$

the mean reputation has fixed point

$$m^*(p) = \frac{p\alpha_+}{p\alpha_+ + (1 - p)\alpha_-}. \quad (47)$$

Writing $\rho = \alpha_-/\alpha_+$,

$$m^*(p) \geq \frac{1}{2} \iff \frac{p}{1 - p} \geq \rho. \quad (48)$$

If trusting an incorrect report costs L_+ and rejecting a correct report costs L_- , Bayes decision theory gives

$$\text{trust} \iff \frac{p}{1 - p} \geq \frac{L_+}{L_-}. \quad (49)$$

The stationary-mean reputation threshold therefore implements the Bayes boundary for every p if and only if $\rho^* = L_+/L_-$. This theorem calibrates Weight. It is not needed for strict propriety of the reporting score in Commit.

Scope of the architecture refinement result

The proof of Theorem 5 is layerwise. Restrict follows from Theorem 4. The preserved parent interface and unchanged local error semantics leave Weight inside the same evaluator-loss calibration problem. Reapplying the same parent-facing reporting contract in the same strategic scope preserves Commit, including the unchanged report aggregator. The preserved interface and parent context determine the same type of local utility contribution, so the parent continues to apply its Aggregate operator. Internal nodes of the inserted subtree retain their own locally valid contracts, and unaffected nodes are unchanged. Induction gives the same result after repeated conservative refinement.

Data and code availability

The general results are analytical and no new empirical dataset is introduced in this synthesis. The two figures are schematic theorem illustrations. The source package includes the plotting scripts used to generate them. Any component-specific empirical results mentioned for the hierarchical one-bit evaluation study, the categorical allocation-verification study, and the multi-resolution attribution study are reported in the cited component papers.

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